

SIDDHARTH YADAV

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EDUCATION

VIT Bhopal University

Bhopal, Madhya Pradesh, India

Bachelor of Technology (B.Tech), Health Informatics

Currently in 7th Semester

- Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Artificial Intelligence, Machine Learning, Data Analytics, Healthcare Informatics, Electronic Health Records, Statistics/Probability, Python Programming

TECHNICAL SKILLS

Programming: Python, C++, SQL

Data Science & Analytics: Pandas, NumPy, Data Cleaning, Exploratory Data Analysis, Data Visualization, Statistical Analysis

Machine Learning: Classification, Predictive Modeling, Random Forest, XGBoost, Model Evaluation, SHAP / Explainable AI

Databases & Visualization: SQL, Relational Databases, Power BI, Dashboard Development

AI / Development: LLM APIs (Groq), Pydantic, Prompt Engineering, RAG (Retrieval-Augmented Generation), FastAPI, Streamlit, AI Application Development

Healthcare Domain: Health Informatics, EHR Concepts, Clinical/Healthcare Analytics

Data Structures & Algorithms: Arrays, Strings, Hashing, Sliding Window, Two Pointers, Prefix Sum, Binary Search, Linked Lists, Heaps, Graph Fundamentals, Recursion/Backtracking

Soft Skills: Communication, Analytical Thinking, Problem-Solving, Teamwork & Collaboration, Time Management, Adaptability

INTERNSHIP EXPERIENCE

Intern, Hindustan Aeronautics Limited (HAL)

May 2026 – Jul 2026

- Gained hands-on exposure to a professional engineering organization, working within established organizational workflows and processes.
- Collaborated on practical, real-world tasks in a professional technical environment, strengthening problem-solving and workplace-readiness skills.

PROJECTS

Healthcare Patient Readmission Prediction

Python, SQL, ML, Power BI, Streamlit

GitHub: github.com/siddharthjunior7-ux/healthcare-readmission-prediction

- Built an end-to-end machine learning pipeline to predict 30-day hospital readmission risk for diabetic patients using the Diabetes 130-US Hospitals dataset.
- Performed data cleaning, exploratory data analysis, and feature engineering in Python/Pandas, and used SQL to extract and analyze healthcare data insights.
- Trained and evaluated classification models (Random Forest, XGBoost) for readmission prediction and applied SHAP for model interpretability.
- Designed a Power BI dashboard concept and a Streamlit interface to display patient-level risk predictions and key contributing factors.

AI Resume-Job Fit Analyzer

Python, Groq API (LLM), Pydantic

- Built an AI-driven application that compares a candidate's resume against a job description to evaluate alignment.
- Parsed resume content (PDF/DOCX) and used Pydantic schemas with an LLM (Groq API) to structure the output into a fit percentage and match verdict.
- Designed the tool to surface relevant skill matches and gaps, improving understanding of resume-to-job alignment.

AI Property Q&A Agent (Real Estate)

Python, Groq API (LLM), Pydantic, RAG

GitHub: github.com/siddharthjunior7-ux/realestate-property-qa-agent

- Built an AI agent that answers natural-language questions about a real estate property — price, location, amenities, configuration, possession date, and nearby facilities — using an LLM (Groq API).
- Applied retrieval-augmented prompting to ground responses in the source property document, with guardrails that decline to answer when requested information isn't present in the data.
- Adapted a FastAPI + Pydantic architecture from a prior LLM-based Q&A project to structure and validate agent responses.

CERTIFICATIONS

- Applied Machine Learning — University of Michigan (Coursera)
- Generative AI using IBM watsonx — IBM
- Summer Training: Deep Learning and OpenCV — Hindustan Aeronautics Limited (HAL)
- Current Requirements for Conducting Clinical Trials in India for Investigational New Drugs/New Drug — NPTEL

ACHIEVEMENTS / ADDITIONAL

- Independently designed and built three end-to-end technical projects spanning healthcare machine learning, AI/LLM application development, and AI agent development — maintaining all work on GitHub with version control.